IBEHS 5P06: Design History File (DHF)

Weight: 10% of the final course grade
Assessment schedule: Five reviews across the course, normally worth 2% each
Submission location: The designated iBioMed Capstone Microsoft Teams Private Channel

The Design History File (DHF) is the authoritative record of your team's engineering design process. It must allow an informed reader to understand how the project evolved, why decisions were made, and how the current design could be reproduced or continued. The DHF is an actively maintained engineering record and should not just be a folder assembled shortly before a review.

Repository requirements

- Use the course DHF template as the starting structure or an alternative structure if it better suits your project.
- Ensure the teaching team has access to the complete repository before every scheduled review.

1. **Core expectations**
   - Maintain the DHF throughout the project and update it promptly after meetings, design work, analyses, testing, and decisions.
   - Organize files using a clear, consistent structure. You are encouraged to begin with the DHF template posted on Avenue and adapt it to suit the project.
   - Use meaningful file names, dates, revision identifiers, and short explanatory notes so that the purpose and status of each item are clear.
   - Preserve earlier iterations. Do not overwrite the design history in a way that hides changes or abandoned approaches.
2. **Evidence to include**
   - Stakeholder communications, user research, background literature, standards, codes, regulatory resources, and other sources that shaped the project.
   - Meeting agendas, minutes, action items, discussion documents, decisions, feedback, and records of how feedback was addressed. Record roles and responsibilities of each group member and tasks each member completed.
   - Design inputs and outputs, concept-generation records, evaluation matrices, calculations, models, simulations, risk analyses, FMEAs or approved equivalents, and compliance plans.
   - Drawings, schematics, CAD files, code, pseudocode, bills of materials, manufacturing information, data, test procedures, raw and processed results, photographs, and videos.
   - Records of failed approaches, unexpected results, changes in scope, and the reasoning behind major revisions.
   - Enough context and explanation for another technically informed person to interpret and, where appropriate, reproduce the work.
3. **Project planning and progression**
   - Maintain a current Gantt chart showing tasks, owners, timing, resources, dependencies, technical contributions, implementation, and verification activities.
   - Create dated revisions rather than replacing prior versions. Explain significant differences between planned and actual progress.
   - Make the contributions and responsibilities of each discipline and team member visible.
4. **Design-review documentation**
   - Maintain a discussion document in the DHF's Design Review area that summarizes the current design state, open questions, recent changes, and plans for the next stage.
   - Update the discussion document before instructor meetings so instructors can focus on review and guidance rather than reconstructing the project history.

- Record the decisions, feedback, and action items arising from each review and show how the team responded.

5. Review and grading

- Teaching assistants will review the DHF twice in the Fall term, twice in the Winter term, and once at the end of the course.
- Evaluation will consider completeness, organization, traceability, evidence of iteration, reproducibility, currency, and the quality of the engineering record.
- A large number of files is not sufficient. Strong DHFs explain how evidence relates to requirements, decisions, risks, implementation, and verification.
- The instructional team must be able to access all submitted files and links at the time of review.

DHF checklist

- The current design state and next steps are easy to find.
- Important decisions can be traced to evidence, analysis, stakeholder input, or instructor feedback.
- Superseded and failed approaches remain documented with an explanation.
- The Gantt chart and discussion document are current.
- Technical files open correctly and are accompanied by enough context to be understood.
- Individual and discipline-specific contributions are visible.
- Use a clear folder structure, consistent file names, dates, and revision identifiers.
- Confirm that all links and files are accessible to the teaching team and open correctly.
- Keep essential engineering records in editable or discipline-appropriate native formats where practical.
- Do not place confidential stakeholder or personal information in the DHF unless its collection and storage have been authorized.
- Document and cite any permitted preliminary use of generative AI. Generative AI may not be used to produce final technical submissions or written deliverable content.
- Maintain the DHF in the designated Microsoft Teams repository.